Superstore Sales Data Analysis: Final Report

Internova Data Analytics Virtual Internship | Prepared by Jair Alejandro Forero Rodríguez | November 2026

1. Executive Summary

This report analyzes retail superstore sales from January 2015 to December 2018 using Python, SQL, and Power BI. Revenue grew from $479,575 in 2015 to $722,052 in 2018 (+50.6%, a compound growth of 14.6% per year), reaching $2.26 million in total across 4,922 orders and 793 customers.

Main findings:

- Growth came from customers ordering more often, not from larger orders: average order value (AOV) fell 14.2% while orders per active customer rose 49.7%.
- Sales are strongly seasonal: September, November, and December generate about 43% of average annual revenue.
- Revenue is concentrated: California and New York generate a third of total revenue, and the West region's lead is mostly a California effect.
- The South is the weakest region, with the lowest revenue share and the slowest growth.
- Machines and Copiers produce 14.8% of revenue from only 1.9% of order lines, which makes revenue sensitive to a few large sales.

The dataset contains only Sales (no profit, cost, quantity, or discount), so every conclusion refers to revenue, not profitability.

2. Dataset and Scope

Item	Detail
Source	Internova virtual internship, Superstore sales dataset
Records	9,800 raw, 9,799 after cleaning
Columns	18 original and 6 derived
Period	3 January 2015 to 30 December 2018

Item	Detail
Granularity	One row is one product line within an order
Geography	United States: 4 regions and 49 states (including the District of Columbia)
Products	3 categories, 17 sub-categories, 1,861 products

3. Method and Data Cleaning

The data was cleaned and explored with Python (pandas, matplotlib, seaborn). The same questions were then answered with SQL (SQLite) and the results were validated against pandas. Finally, an interactive Power BI dashboard was built.

Step	Issue found	Action taken	Result
Data types	Order Date and Ship Date loaded as text; Postal Code as a number	Dates converted to datetime; Postal Code converted to text	Correct data types
Missing values	11 records (0.11%) without Postal Code, all in Burlington, VT	Filled with 05401 as a documented assumption, since no other Burlington record had a code	0 missing values; no impact on region, state, or city analysis
Duplicates	1 record identical to another except Row ID	Removed (impact $281.37, 0.0124% of revenue)	9,800 to 9,799 records
Categorical values	Checked for typos, capitalization, and spaces	None found	No changes needed
Date consistency	Checked date order, ranges, and shipping times	No Ship Date before Order Date; shipping takes 0 to 7 days and matches each ship mode	No changes needed

Step	Issue found	Action taken	Result
Derived columns	Needed for time analysis	Created Shipping Days, Order Year, Month, Month Name, Quarter, and Weekday	24 columns

4. Key Metrics

Metric	Value
Total revenue	$2,261,255.41
Unique orders	4,922
Unique customers	793
Unique products	1,861
Average order value (AOV)	$459.42
Average lines per order	1.99
Average orders per customer	6.21
Mean vs median sales per line	$230.76 vs $54.48 (strongly right-skewed)
Average shipping time	3.96 days

Year	Revenue	Orders	Active customers	AOV	Revenue growth
2015	$479,574.84	947	589	$506.41	-
2016	$459,436.01	1,019	567	$450.87	-4.20%
2017	$600,192.55	1,295	635	$463.47	+30.64%
2018	$722,052.02	1,661	690	$434.71	+20.30%

5. Insights and Recommendations

Insight 1: Revenue grows, but the average order is getting smaller

Evidence: Between 2015 and 2018, revenue grew 50.6% and orders grew 75.4%, while AOV fell 14.2% ($506.41 to $434.71). Lines per order fell from 2.06 to 1.96, and average sales per line fell 9.8% ($245.68 to $221.62), while the median per line stayed stable (about $52 to $54). Typical sales did not get cheaper; the average fell because of fewer very large sales (Machines declined 29.8%). The 2016 dip (-4.2%) was driven largely by Machines, which fell $34,260, more than the entire net decline of $20,139.

Recommendation: Increase basket size with bundles and cross-selling, and track high-ticket sales separately from everyday sales so a few large deals do not hide the trend.

Insight 2: Sales are highly seasonal

Evidence: September ($75,026), November ($87,540), and December ($80,370) are the strongest months on average and account for about 43% of average annual revenue. Q4 alone generates 37% to 39% of revenue every year. January ($23,573) and February ($14,843) are the weakest months. The pattern repeats in all four years.

Recommendation: Plan inventory, staffing, and promotions ahead of September to December, and use the low months for targeted campaigns to even out demand.

Insight 3: Growth comes from order frequency and loyal customers, not from more customers

Evidence: Active customers grew 17.1% (589 to 690), but orders per active customer grew 49.7% (1.61 to 2.41), and revenue per active customer rose from $814 to $1,046. Customers with 6 or more orders in a year went from 1 in each of 2015 and 2016 to 13 in 2017 and 12 in 2018. Earlier customer cohorts stayed active at above 80% in 2018.

Recommendation: Protect repeat buyers with loyalty or reorder incentives, and reactivate inactive customers before the September to December peak. The data cannot show what caused the higher frequency (promotions, new products, or campaigns), so campaign and discount data should be recorded to test it.

Insight 4: Revenue depends on a few markets

Evidence: California ($446,306, 19.7%) and New York ($306,361, 13.6%) together generate 33.3% of revenue; the top 5 states generate 51.9%, and 16 of 49 states generate 80%. New York City ($252,463, 11.2%) is the top city, and the top 10 cities generate 44.3% of revenue. The West leads by region (31.4%), but California alone is 62.8% of West revenue, and the West without California would be smaller than the South. California leads through order volume (1,002 orders versus 547 for New York), not order size (AOV $445 versus $560).

Recommendation: Study California and New York City as markets (customer base, order frequency, campaigns) instead of copying a regional strategy, adjust the analysis for population to see whether other large metro areas are underserved, and track concentration as a business risk.

Insight 5: The South is the weakest region

Evidence: The South has the lowest revenue share (17.2%) and the slowest growth between 2015 and 2018 (+18.2%, versus +70.1% for the West, +65.0% for the East, and +37.6% for the Central region). Its revenue fell 32.2% in 2016, and its share dropped from 21.6% to 16.9%. The Central region lost share in 2018 (revenue -2.8%). Customer segment mix (about 50% Consumer, 30% Corporate, 19% Home Office) and shipping times (3.9 to 4.1 days) are almost identical across regions, so they do not explain the gap.

Recommendation: Review the South market and set a growth target, comparing its customer base and product mix with the East, which grew steadily every year.

Insight 6: Product revenue is balanced by category but concentrated in a few sub-categories and products

Evidence: Technology (36.6%), Furniture (32.2%), and Office Supplies (31.2%) are evenly balanced. Phones and Chairs alone generate 28.8% of revenue, 9 of 17 sub-categories generate 83.4%, and the top 10% of products generate 59.1%. Machines and Copiers represent 14.8% of revenue from 1.9% of lines, and the Canon imageCLASS 2200 copier alone accounts for 42% of Copiers revenue. Furniture lost share (32.6% to 29.4%), with growth slowing to +8.4% in 2018, while Office Supplies gained share.

Recommendation: Protect the leading sub-categories, treat Machines and Copiers as high-value but volatile, and review why Furniture growth is slowing.

6. Dashboard

An interactive Power BI dashboard supports these findings. The overview page shows revenue, unique orders, customers, AOV, and 2015 to 2018 growth, together with revenue by sub-category, revenue by state, and monthly revenue by year. Detail pages cover customer behavior and revenue drivers, and geography and products. Filters allow the analysis by year, category, and location.

7. Limitations and Assumptions

- The dataset has no profit, cost, quantity, or discount columns, so rankings are by revenue, not profitability, and price effects cannot be separated from quantity or discounts.
- Findings describe what happened, not why. Explanations such as promotions or campaigns are hypotheses.
- Customers are counted by delivery location, and customers ship to several regions, so customer counts by region are not unique customers.
- Customers first seen in 2015 may have bought before the dataset starts, so new means new to this dataset.
- Revenue by state is not adjusted for population or market size.
- The 11 missing Postal Codes were filled with an assumed value (05401) and one duplicate record was removed; neither affects the conclusions.

8. Conclusion and Priority Actions

The business is growing at 14.6% per year, but growth relies on frequent smaller orders, a few strong months, and a few large markets. The priority actions are:

1. Increase average order value through bundles and cross-selling.
2. Prepare for the September to December peak and build demand in the weak months.
3. Strengthen loyalty and reactivation programs for repeat customers.
4. Reduce dependence on California and New York, and review the South strategy.
5. Start recording cost, quantity, discount, and campaign data, so the next analysis can measure profitability and explain the causes.